

EXPECTATIONS AS ENDOWMENTS: EVIDENCE ON REFERENCE-DEPENDENT PREFERENCES FROM EXCHANGE AND VALUATION EXPERIMENTS*

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While evidence suggests that people evaluate outcomes with respect to reference points, little is known about what determines them. We conduct two experiments that show that reference points are determined, at least in part, by expectations. In an exchange experiment, we endow subjects with an item and randomize the probability they will be allowed to trade. Subjects that are less likely to be able to trade are more likely to choose to keep their item. In a valuation experiment, we randomly assign subjects a high or low probability of obtaining an item and elicit their willingness-to-accept for it. The high probability treatment increases valuation of the item by 20–30%. *JEL Codes:* C91, D03, D11, D81, D84.

I. INTRODUCTION

Evidence from a variety of settings indicates that people are loss averse: they dislike losses much more than they enjoy equal-sized gains. Yet little is known about the determination of the reference points relative to which gains and losses are defined. Kahneman and Tversky's (Kahneman and Tversky 1979; Tversky and Kahneman 1991) highly influential prospect theory, where loss aversion was first introduced, left the reference point imprecise. It has often been taken to be the status quo (Samuelson and Zeckhauser 1988), meaning that individuals are reluctant to give up things they currently possess. An alternative view, advocated by Köszegi and Rabin (2006, 2007, 2009) (henceforth KR), is that reference points are determined by recent expectations about outcomes, which need not correspond to the status quo. Because the status quo often determines expectations, much existing evidence is consistent with either alternative. However, the implications of loss aversion frequently differ depending on the assumed

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reference points. This study provides new experimental evidence that reference points are determined, at least in part, by expectations: people appear loss averse around their expected outcomes.

We present two experiments in which we manipulate expectations separately from endowments. Our first experiment tests whether expectations affect exchange behavior. We endow all subjects with the same item (a mug) and randomize the probability that they will be allowed to exchange that item for an alternative (a pen). We find that subjects that are more likely to *expect* to keep their endowed item (because they have a low probability of being allowed to exchange) are more likely to *choose* to keep their item if given the opportunity to exchange. This finding is predicted if individuals have expectation-based reference points, but not by theories that identify the reference point with the status quo or by theories in which preferences do not depend on reference points. It suggests that the expectation of continued ownership, rather than current formal ownership per se, induces a reluctance to part with possessions (a phenomenon Thaler [1980] called the “endowment effect”). As a consequence, loss aversion need not impede transactions that individuals expect to occur.¹

Our second experiment measures the effect of expectations on subjects’ monetary valuation of an item. Subjects are not initially endowed with any item but learn that they will receive a mug for free with either high (80%) or low (10%) probability. All subjects also know that with probability 10%, they instead have a choice between the mug and money. We elicit their mug/money choices for varying amounts of money, which gives a measure of their willingness-to-accept (WTA) for the mug. To increase the precision of our estimated treatment effect, we also elicit WTA for an unrelated item, a pen. This allows us to proxy for idiosyncratic factors that affect valuation of university merchandise.² We estimate that subjects who were randomly assigned to have a high expectation of leaving with the mug value the mug about 20–30% higher than subjects who were less likely to leave with the mug, an effect that is both statistically and economically significant.

1. As discussed by DellaVigna (2009), this argument could explain List’s (2003) finding that experienced sports card traders, who may expect to trade the item they get endowed with, do not seem to display the endowment effect. Engelmann and Hollard (2010) instead argue that trading experience reduces “trade uncertainty” (the perception of how risky the act of trading is) and that trade uncertainty is a source of observed endowment effects.

2. Smith (2008) conducts a similar study but does not proxy for such idiosyncratic factors. His experiment is discussed in more detail in Section III.A.

Moreover, we also conduct a variation of our second experiment to untangle whether expectations matter because people experience gains and losses with respect to a reference point, or because likely ownership of an item increases individuals' estimates of its consumption utility. According to the former theory, expectations should only affect the valuation of the mug currently in a person's reference point. The latter theory, which we refer to as "motivated taste change," instead predicts that expecting to own one mug should also increase the valuation of a second, identical mug. We do not find any evidence that motivated taste change is the source of our results: subjects' expectations of getting one mug do not affect their valuation of a second mug conditional on getting the first mug.

Expectation-based loss aversion has various implications for consumer behavior and policy. As an example, imagine a potential car buyer who forms an expectation that she will buy a particular model. Loss aversion may lead the buyer not to cancel her purchase when she learns at the dealership that the car gets less than 20 miles per gallon or that the financing options are less advantageous than she had thought, even though if she had known this information from the beginning she would not have planned on buying that car. Thus, the timing of such disclosures can be very important for how they affect demand when expectations determine reference points, unlike when people do not display loss aversion or are loss averse with respect to the status quo. More generally, firms' pricing and marketing strategies will attempt to instill in consumers an expectation of purchase, which may increase their willingness to pay.³ Policy makers also benefit from "expectations management," as they may face less resistance against a policy change if it was expected than if it comes as a surprise.

There is little other experimental research that directly examines the effect of expectations on reference points.⁴ [Knetsch](#)

3. The theoretical implications of expectation-based reference points for optimal pricing strategies are studied by [Heidhues and Köszegi \(2008, 2010\)](#) and [Herweg \(2010\)](#). Other recent theoretical work applies the KR model to study optimal incentive contracts ([Herweg, Müller, and Weinschenk 2010](#); [Macera 2010](#)) or bidding behavior in auctions ([Lange and Ratan 2010](#)).

4. Some recent field evidence on cab drivers' labor supply ([Crawford and Meng 2011](#)), effort choice of professional golf players ([Pope and Schweitzer 2011](#)), risky choices in "Deal or No Deal" ([Post et al. 2008](#)), domestic violence ([Card and Dahl 2011](#)), and police performance after final offer arbitration ([Mas 2006](#)) is also consistent with reference points being determined by expectations.

and Wong (2009) conduct exchange experiments using a variety of procedures and discuss their results in terms of the KR theory. They do not manipulate expectations explicitly as we do in our first experiment and do not directly test the theory of expectation-based reference points. Loewenstein and Adler (1995) examine how probabilistic endowment affects subjects' valuation of an item; however, because expectations are endogenous to plans, it is ambiguous how their manipulation affects expectation-based reference points.⁵ Another study that exogenously manipulates expectations and finds support for expectation-based reference points is Abeler et al. (2011). They conduct a real-effort experiment in which subjects are equally likely to receive a fixed payment or to be paid according to their effort, and only learn which case applies after they stop working. When the amount of the fixed payment is higher, subjects work more. This is predicted when expectations determine reference points, as subjects want to avoid being disappointed by potentially earning less than their fixed payment.⁶ In this study, subjects are not informed that the level of the fixed payment (high/low) is randomly assigned. As a consequence, subjects may make (potentially mistaken) inferences from the amount of the fixed payment about the "appropriate" amount of effort to provide, and this could contribute to the observed treatment effect (e.g., if the fixed payment is perceived as an indication of how much the experimenter expects participants to work).⁷ We provide evidence that transparent randomization can be important: in pilot sessions of our first experiment, subjects were not aware that their probability of being allowed to exchange was randomly determined, and they made inferences about the relative value of the items, which then influenced their behavior.

5. In their Experiment 2, Loewenstein and Adler compare WTA for a mug between two groups: (1) subjects who lost a coin flip and expect to not get a mug, and (2) subjects who face a coin flip in which they could either get nothing or get a mug with the option to trade it for money. They find little difference in average WTA of the two groups. Because subjects in group 2 may have planned on trading the mug for money, their expectation of leaving with the mug is fully endogenous to their choices. As a consequence, the KR theory does not make an unambiguous prediction here. In contrast, in our second experiment, we exogenously manipulate subjects' expectations, assigning them the mug with either 10% or 80% probability without an opportunity to trade it.

6. See also a related study by Gill and Prowse (forthcoming) who find evidence for disappointment aversion in a two-person sequential-move tournament.

7. The concern that subjects might conform to their interpretation of appropriate effort is distinct from anchoring or gift-exchange motives, which Abeler et al. control for and find no evidence of in additional treatments.

Our two experiments make randomization transparent, and these inferences disappear. We thus confirm that expectations affect reference points, and furthermore provide new evidence that expectations matter not only for effort provision but also for exchange behavior and the valuation of goods.

II. EXPERIMENT 1: EXPECTATIONS AND EXCHANGE BEHAVIOR

In our first experiment, we endow subjects with an item (a university travel mug) and randomly and transparently manipulate the probability with which a subject will have the opportunity to exchange her item for an alternative (a silver metal university pen).⁸ We first derive the theoretical predictions, then describe our experimental procedures, and present the results.

II.A. *Theoretical Predictions for Behavior*

In this section, we compare how expectations affect the exchange behavior of three types of individuals: an individual with classical preferences (no loss aversion), one with loss aversion around a reference point given by the status quo, and one with loss aversion around a reference point determined by expectations (as in KR 2006).⁹ Consider an individual who is endowed with a mug and expects to have the option to exchange this mug for a pen with probability p . After being given some time to think about the situation, she is asked to register a decision: if the option to exchange materializes, would she like to exchange?

Let individuals have utility functions $u(c|r)$ that can depend both on consumption c and reference levels r . Consumption and reference levels have multiple dimensions; for this setting, only the “mug” dimension and the “pen” dimension are relevant. Utility on the k th dimension is composed of direct consumption utility u_k and gain-loss utility with respect to that dimension’s reference level of utility u_k^r . Total utility from a consumption outcome c is then given by:

8. Both items have a retail value of around \$8. In a typical endowment effect experiment (such as Knetsch 1989), half the subjects would be endowed with a mug and the other half with a pen. We chose to endow all our subjects with the same item (which we randomly picked by flipping a coin before our first session) to get sufficient power for testing the hypothesis that expectations affect exchange behavior.

9. A more detailed exposition of the KR theory is provided in the Appendix.

$$u(c|r) = \sum_{k \in \{mug, pen\}} \left\{ \underbrace{u_k}_{\text{Consumption utility}} + \underbrace{\mu(u_k - u_k^r)}_{\text{Gain-loss utility}} \right\},$$

where μ has the properties of the Kahneman-Tversky value function. For simplicity, we follow Section IV of KR (2006) and assume μ to be a piecewise linear function with a kink at 0 that captures loss aversion: $\mu(x) = \eta x$ for $x > 0$ and $\mu(x) = \eta \lambda x$ for $x \leq 0$, where $\eta \geq 0$ is the weight on gain-loss utility and $\lambda > 1$ is the individual's loss aversion coefficient. This specification nests classical preferences that do not feature gain-loss utility ($\eta = 0$). Throughout, when referring to individuals with reference-dependent preferences, we assume $\eta > 0$.

Now we consider two different specifications of the reference point. If the reference point is given by the status quo, which is that the individual owns the mug but not the pen, then we simply have $u_{mug}^r = u_{mug}$ and $u_{pen}^r = 0$.

In contrast, in KR's specification, the reference point is determined by the individual's probabilistic beliefs about outcomes, in other words, her expectations. KR assume that individuals calculate an outcome's gain-loss utility as follows: they compare each dimension's consumption utility from the actual outcome with that of each possible outcome, producing a series of gain-loss $\mu(\cdot)$ terms. Each possible outcome's $\mu(\cdot)$ is then weighted by the probability with which the individual expected that outcome to occur.¹⁰ Furthermore, an individual's reference point is *endogenous* to her plans, as the probabilities of different states of the world occurring may depend at least in part on her decisions.

In our example, if the KR individual plans not to exchange, then her reference point is the same as in the status quo case—she keeps the mug for sure. If instead she plans to exchange conditional on having the possibility to do so, the two states of the world she considers for her gain-loss utility are $r_{end\ w/pen} = \{1\ \text{pen}, 0\ \text{mugs}\}$ and $r_{end\ w/mug} = \{0\ \text{pens}, 1\ \text{mug}\}$, which occur with probability p and $1 - p$, respectively. Her utility of consumption bundle c given her expectations is then $pu(c|r_{end\ w/pen}) + (1 - p)u(c|r_{end\ w/mug})$. If the exchange occurs, she gains or loses nothing with respect to $r_{end\ w/pen}$, but with respect to $r_{end\ w/mug}$, she

10. In this application, no predictions would change if reference points were instead simply given by the expected consumption utility in each dimension, in the spirit of Bell (1985).

gains u_{pen} and loses u_{mug} . Thus, her total utility of this outcome would be her consumption utility u_{pen} plus her gain-loss utility $(1 - p)\eta(u_{pen} - \lambda u_{mug})$. Similarly, if the exchange does not occur, her total utility would be $u_{mug} + p\eta(u_{mug} - \lambda u_{pen})$. Following KR, we assume that the individual acts to maximize her expected utility given her reference point (which itself is determined by her plan) and chooses the plan that leads to the highest expected utility, given her rationally forecasted future actions.¹¹

Now, let individuals vary in their value for the mug and the pen; assume that (u_{mug}, u_{pen}) are distributed according to some population distribution function. The following proposition shows that increasing the probability p with which an individual is allowed to exchange increases the expected proportion of individuals who choose to exchange when reference points are determined by expectations, but not when reference points are given by the status quo or when individuals do not have gain-loss utility.

PROPOSITION 1. If reference points are given by the status quo or otherwise unaffected by expectations, or if there is no gain-loss utility ($\eta = 0$), the probability p of having the option to exchange does not affect the decision to exchange. If reference points are determined by expectations, then increasing p increases the expected proportion of individuals who choose to exchange.

Proof. When reference points are given by the status quo, the expected utility from choosing to exchange is greater than the utility of keeping the mug if and only if $p[u_{pen} + \eta(u_{pen} - \lambda u_{mug})] + (1 - p)u_{mug} > u_{mug}$, which holds if and only if $u_{pen} > \frac{1+\eta\lambda}{1+\eta}u_{mug}$ and thus does not depend on p . A similar condition, not dependent on p , holds for any fixed reference point. When $\eta = 0$, the individual strictly desires to exchange if and only if $u_{pen} > u_{mug}$.

However, when reference points are determined by expectations, an individual desires to exchange if and only if the expected utility of planning to exchange (and following through on the plan) is greater than the expected utility of planning to keep the mug (and keeping it). As the Online Appendix shows, this is

11. In the language of KR, we assume that the individual plays the “preferred personal equilibrium.” However, the theoretical prediction on the effect of p is the same when using the less restrictive “personal equilibrium” concept, or Kőszegi and Rabin’s (2007) “choice-acclimating personal equilibrium,” which assumes that reference points are determined by the individual’s decision, not her plan. See the Online Appendix for more details.

true when $(u_{pen} - u_{mug}) + \eta(1-p)(1-\lambda)(u_{pen} + u_{mug}) > 0$. Note that $u_{pen} > u_{mug}$ is a necessary condition for the individual to desire to exchange, as for $p < 1$ the second term is negative due to loss aversion. The second term gets less negative as p increases. Therefore, an individual who chooses to keep the mug under one value of p will always do so for lower values of p , but may instead choose to exchange for higher values of p . ■

Intuitively, for an individual who has classical preferences, the decision of whether to exchange is influenced only by the consumption utilities she derives from the two items. On the other hand, an individual who is loss averse around the status quo may not be willing to give up the mug he owns for the pen, even if the pen has higher consumption utility, due to the loss he feels from moving away from the status quo. However, this loss, and therefore the decision, is independent of whether the chance of the exchange occurring is high or low.

Meanwhile, for an individual with KR preferences, the cost imposed by the gain-loss utility in case she decides to exchange must be outweighed by the gain in consumption utility from getting the pen instead of the mug, and this is less likely to happen the lower p is. On the other hand, as $p \rightarrow 1$, gain-loss utility becomes relatively less important. In the extreme, if such an individual were certain to be able to exchange, she would prefer to do so if and only if the consumption utility from the pen exceeds the one from the mug, exactly like a classical individual.

II.B. Procedures

On arrival at the lab, each subject is seated at a carrel with a mug and a pen on it.¹² We then flip a coin in front of each subject individually. The coin's sides are labeled 1 and 9, and we give each subject an index card with their resulting number on it. Subjects then start reading instructions on their computer screen. Subjects are told that they own the mug in front of them and that each participant will leave the experiment with either a mug or a pen. They are informed that at the end of the session, they may have the option to exchange their mug for the pen, if they so desire, and this option will occur if and only if a 10-sided die that we roll individually for each subject at the end of the study comes up lower than or equal to the number on their index card. Thus, subjects whose coin came up 1 have a 10% chance of having the

12. More details on both experiments are provided in the Online Appendix.

option to exchange (we refer to them as being in *Treatment T^L* , for low probability of being able to trade), whereas subjects who got a 9 have a 90% chance of being able to exchange their mug for the pen if they would like (*Treatment T^H*).

After this first round of instructions, and after an experimenter reads the most important parts out loud, subjects answer some demographic questions and fill out the first part of a 44-item “Big Five” personality questionnaire (John, Donahue, and Kentle 1991). The questionnaire allows time for reference points to adapt to subjects’ plans regarding whether to exchange the mug if given the opportunity to do so; in the KR theory, reference points need not instantaneously adapt to changes in expectations but are determined by recent expectations. This questionnaire also serves as a decoy for the main decision we are interested in. After subjects finish answering the first 22 questions, we remind them of the instructions and procedures for the (possible) mug-pen exchange, to make sure they understand and also to make them think about their decision as to whether to exchange the mug. Then, after they answer the second 22 questions, subjects are asked to make a choice *conditional* on the die coming up lower than or equal to the number on their index card.¹³ This allows us to observe a decision for each subject, not just the ones for which the decision actually turns out to apply, which is similar to the strategy method often used in experimental games.¹⁴ Before rolling the die, we ask some additional questions, the answers to which are discussed in the working paper version of this article.

The experiments were run at the Harvard Decision Science Laboratory. A total of 45 subjects (23 females; mean age 21), all of them undergraduate students at the university, participated. We conducted 10 sessions with between three and seven subjects per session. Half the sessions were run on one day in late October

13. They are asked to check one of the following: If the the die comes up [1 in *Treatment T^L* ; 1 to 9 in *Treatment T^H*]: ☐ I want to keep the mug; or ☐ I want to trade the mug for the pen.

14. As is further discussed in the Online Appendix, eliciting a subject’s decision conditional on a choice set being reached is theoretically equivalent to asking the subject to choose once the choice set is reached, if by then the subject has made her plan. This is because KR (2006) posit that reference points are determined by lagged expectations. An interesting avenue for future research would be to test this prediction and investigate how quickly expectation-based reference points adapt. However, if in our experiment we were to only elicit decisions after the uncertainty was resolved, we would need a prohibitively large number of subjects to examine decisions in the 10% possibility-of-trade condition.

2009, the other half over two days in early November. Subjects received a show-up fee of \$10, and the experiment took about 20 minutes.¹⁵

II.C. Results

RESULT 1. Subjects that have a 10% chance of being able to exchange are significantly less likely to be willing to exchange than subjects that have a 90% chance. The proportion of subjects who choose to exchange the mug for the pen is 22.7% in Treatment T^L and 56.5% in Treatment T^H ($p = .033$, two-sided Fisher's exact test).

Thus, we confirm the prediction of KR (2006) that reference points matter for choice and that they are determined by expectations. The effect is large: subjects in Treatment T^H are 34 percentage points, or one and a half times, more likely to choose to exchange than subjects in Treatment T^L . As a test of the robustness of the result, Table I reports the estimated marginal effects from probit regressions that predict the probability a subject chooses to exchange from a treatment indicator and other covariates that may be related to their choice. Among the covariates we consider, gender significantly predicts the desire to exchange: column (2) shows that females seem to like the pen relatively more than males. However, the gender effect does not drive our result, as our treatments were perfectly gender-balanced. Column (3) shows that when both treatment and gender indicators are included as regressors, the indicator for Treatment T^H is still significant at $p < .02$. Furthermore, column (4) shows that adding subject age and an indicator for the day on which the session took place does not change the result either. If anything, the estimated treatment effect becomes even larger.¹⁶

15. In the November sessions, the subjects afterward participated in a second, completely unrelated experiment. The fact that these experiments were unrelated was made very clear to participants, and they were told that the first experiment involved the mugs and pens while the second involved choosing between various payment amounts. The results demonstrate that behavior was similar in the October and November sessions.

16. The robustness of the effect across subsets of sessions is strong: the proportions of subjects in Treatments T^L and T^H who wanted to exchange were 23.1% and 53.9% in our October sessions (13 subjects per treatment) and 22.2% and 60.0% in our November sessions (9 subjects in Treatment T^L , 10 subjects in Treatment T^H).

TABLE I
DETERMINANTS OF DESIRE TO EXCHANGE MUG FOR PEN

Pr(choose to exchange)	(1)	(2)	(3)	(4)
Treatment T^H	0.338** (0.137)		0.353** (0.139)	0.402** (0.154)
Female		0.249* (0.141)	0.270* (0.145)	0.291* (0.163)
Age				0.103 (0.066)
Day indicators	No	No	No	Yes
# observations	45	45	45	45

Notes. Displayed coefficients are predicted marginal effects from probit regressions (in case of dummy variables, for a discrete change from 0 to 1). Dependent variable: Indicator variable = 1 if subject wants to exchange. Standard errors in parentheses.
Level of significance: * $p < .10$, ** $p < .05$, *** $p < .01$

II.D. The Importance of Transparent Randomization

In a pilot for this experiment, we ran sessions in which subjects were simply told that they have a 10% or 90% probability of permitted trade, without being informed that this probability was randomly assigned. In these sessions, the results went in the opposite direction of the ones in our main sessions: subjects with a low probability of being able to exchange were more willing to exchange.¹⁷ After the first few sessions, we realized that this may have been due to a value inference effect: subjects who were given a low (10%) probability of being able to exchange their mug for the pen may have inferred that the pen must therefore be more valuable. We then added a question to the debriefing survey in which we asked subjects which item they believe has higher retail value, and found that indeed the subjects in the low-probability condition were significantly more likely to believe that the pen had higher retail value.

This value inference effect disappears when randomization is made transparent by flipping a coin in front of each subject: the answers to the debriefing question about relative retail value do not differ significantly across treatments in our main sessions. Thus, we conclude that in the pilot, our test of expectation-based reference points was confounded by an experimental design that created subject misconceptions about the relative values of the two items. The pilot may also speak to the relative strength of expectation-based reference points versus value inference: the

17. More detail is provided in the Online Appendix.

effect of expectation-based reference points that we document in our clean treatments can be more than outweighed by (perceived) value signals provided by probabilities. This observation was part of the motivation behind Experiment 2, the goal of which was to get a quantitative measure of the strength of the effect of expectation-based reference points on valuation.

III. EXPERIMENT 2: EXPECTATIONS AND VALUATION

III.A. Theoretical Predictions for Behavior

Effect of Expectations on WTA. We now study how individuals' valuation of an item is affected by their (exogenously manipulated) probability of getting the item. Consider an individual who is told the following: she will receive a mug for free with probability p , which varies between individuals and is either high ($p_H = 0.8$) or low ($p_L = 0.1$). With probability q ($= 0.1$) the individual will receive the opportunity to choose between receiving the mug or various amounts of money. With probability $1 - p - q$, the individual expects to get nothing. Individuals' WTA in case they have the opportunity to trade the mug for money is elicited using an incentive-compatible modified Becker-DeGroot-Marschak (1964) (BDM) mechanism. How do expectations affect WTA under different theories of the reference point?

As before, let consumption utility have multiple dimensions, here a "mug" dimension and a "money" dimension. We parameterize the consumption utility of money as being linear, but this is not crucial. Proposition 2 shows that the probability p of getting the mug for free alters WTA for the mug if reference points are determined by expectations, but has no effect under other theories of the reference point.

PROPOSITION 2. The probability p of getting the mug for free does not affect WTA for the mug if reference points are given by the status quo or otherwise unaffected by expectations, or if there is no gain-loss utility ($\eta = 0$). If reference points are determined by expectations, then increasing the probability p affects WTA for the mug. For any values of the parameters $\eta > 0$ and $\lambda > 1$, there is a q small enough such that increasing p unambiguously increases WTA for the mug.

Proof. If the status quo (no mug, no money) is the reference point, or if $\eta = 0$, then WTA for the mug simply equals the consumption utility of the mug. For any fixed reference point,

WTA may depend on λ and η , but will not depend on p . The Online Appendix shows that with expectation-based reference points and $q = 0$, WTA for the mug is given by $\left(1 + \frac{\eta(\lambda-1)}{1+\eta}p\right) u_{\text{mug}}$, which is increasing in p . The Online Appendix also considers the case when $q > 0$, and shows that a sufficient condition for WTA to be increasing in p is $q < \frac{1+\eta}{2\eta(\lambda-1)}$. ■

Denote by WTA_{mug} an individual's WTA for the mug, which is affected by p under KR preferences. The intuition underlying Proposition 2 is that a higher probability of getting the mug for free increases the weight on the mug in the individual's reference point. However, WTA_{mug} is also affected by q , the probability that the results of the BDM mechanism are implemented, because the endogenous probability of receiving money or a mug in this case affects the reference point. The Appendix shows that WTA_{mug} is unambiguously increasing in p for q small enough; for $q = 0.1$, as in our experiment, it is sufficient that $\lambda \leq 5$. This is a mild restriction, because it is unlikely that losses loom more than five times larger than gains.

Examining the limit as q goes to 0 ($q = 0.1$ is small in our experiment) clarifies the intuition: when $p = 0$, WTA_{mug} simply gives the consumption utility of the mug, u_{mug} . When $p = 1$, getting the mug is the full reference point, and $WTA_{\text{mug}} = \frac{1+\eta\lambda}{1+\eta} u_{\text{mug}}$. For intermediate values of p , we have $WTA_{\text{mug}} = \left(1 + \frac{\eta(\lambda-1)}{1+\eta}p\right) u_{\text{mug}}$. Thus, the effect of changing expectations is proportional to u_{mug} , suggesting the use of the logarithmic functional form to identify the effect of p as a percentage change in WTA.

Improving Statistical Power. Previous studies have found large variance in subjects' WTA or WTP for items similar to university mugs, which might make it difficult to detect even sizable effects of expectations on WTA. For instance, in an experiment similar to the one in this section, Smith (2008) tests whether a higher (lagged) probability of receiving an item increases subjects' valuation of that item (as measured by WTA of subjects that end up receiving the item and WTP of subjects who do not). The data cannot reject the null hypothesis that lagged expectations do not matter for valuation, but standard errors are large enough that there would not have been enough power to detect reasonably sized effects (e.g., a \$1, or 15–25%, change in valuation).¹⁸

18. Another reason no effect is found may be that valuations are elicited after subjects learn whether they receive the item for free, so that the reference point

To improve statistical power, we control for individual-specific valuation of university merchandise by eliciting WTA for an unrelated item, a university pen. Holding expectations constant, WTA for both the mug and pen will be affected by similar factors, such as individual wealth and attachment to the university. Hence, by controlling for WTA for the pen, the unexplained variance in WTA for the mug should be reduced, and a more precise estimate of the effect of expectations may be obtained.

Consider the following situation. After the individual provides her WTA for the mug, but before she learns whether her choice applies, she is given a surprise decision. She is shown a pen and her WTA for the pen, WTA_{pen} , is elicited using the same modified BDM mechanism as for the mug. The pen-related BDM mechanism will be implemented with probability q , in addition to the probability q that the mug-related BDM mechanism will be implemented.¹⁹ Proposition 3 verifies that WTA_{pen} is unaffected by p for theories of the reference point that do not depend on expectations, and that if a higher p increases WTA_{mug} , then WTA_{pen} will not decrease in p . Thus, controlling for individual WTA_{pen} will not lead to a predicted effect of p on WTA_{mug} when there is no such effect.

PROPOSITION 3. The probability p of getting the mug for free has no effect on WTA for the pen if reference points are given by the status quo or otherwise unaffected by expectations, or if there is no gain-loss utility. If reference points are determined by expectations and WTA for the mug is increasing in p , then WTA for the pen will weakly increase in p .

may already have adjusted to having the item (or not having it) for sure. Indeed, an experiment by Loewenstein and Adler (1995) suggests that reference points in such experiments begin to adapt quickly. They examine subjects who may receive a mug as a result of a coin toss (50% probability) and are asked to state their WTA for the mug contingent on receiving it. After the coin flip, subjects who correctly guessed the coin toss are given the mug and have the opportunity to revise their WTA (though only hypothetically). On average, this results in a 13% increase in WTA. This is consistent with the reference point quickly adapting to changes in either subjects' ownership of the mug or expectations of getting the mug.

19. Thus, there are four possible states of the world: with probability p , the individual receives the mug independently of any decisions made; with probability q she receives one of her choices between the mug and money; with probability q she receives one of her choices between the pen and money; and with probability $1 - p - 2q$ she receives nothing.

Proof. As in Proposition 2, if the reference point is the status quo (no mug, no money, no pen), then WTA for the pen simply gives the utility of the pen. For any fixed reference point, WTA may depend on λ and η , but will not depend on p .

Now suppose reference points are determined by expectations. If mugs and pens are the same dimension of utility, then the derivation is the same as for Proposition 2, and raising p increases WTA_{pen} . If mugs and pens are independent dimensions of utility, then when $q = 0$, the utility of getting the pen is $u_{pen} + \eta(u_{pen} - p\lambda u_{mug})$ and that of getting \$ WTA_{pen} is $WTA_{pen} + \eta(WTA_{pen} - p\lambda u_{mug})$. At the indifference point, these expressions must be equal; p cancels and WTA_{pen} is independent of the treatment. The Online Appendix takes the case where $q > 0$ and shows that WTA_{pen} does not depend directly on p but does depend positively on WTA_{mug} , as WTA_{mug} will affect the amounts of money the individual expects to receive. This effect is proportional to q . We use the implicit function theorem to show that $\frac{dWTA_{pen}}{dWTA_{mug}} > 0$ for interior values of WTA_{pen} , WTA_{mug} . ■

The proposition shows that there is a trade-off: whereas controlling for WTA_{pen} would not give us a false positive (a positive estimate when the true effect is 0), doing so might attenuate our estimate of the effect of expectations on WTA_{mug} . If reference points are determined by expectations, WTA_{pen} may be affected by p for two reasons. First, mugs and pens may overlap in dimensions of utility, if for instance the relevant utility dimension were “university merchandise.” In this case, p will have a similar effect on WTA_{pen} and WTA_{mug} . Second, p may affect WTA_{pen} indirectly: because p affects WTA_{mug} it alters individuals’ expectations of how much money they will receive. This indirect effect on WTA_{pen} is of the same sign as the effect on WTA_{mug} , and is proportional to q , which is small in our context and thus likely undetectable.²⁰

Our results that follow show that controlling for WTA_{pen} increases the precision of our treatment effect on WTA_{mug} without reducing the point estimate. We cannot statistically reject a zero effect of p on WTA_{pen} , consistent with “mug” and “pen” being in different utility dimensions and the indirect effect through the expected money receipt being small (due to q being low).

20. Another way WTA_{mug} could affect WTA_{pen} is via anchoring, which would raise WTA_{pen} when p is high and also attenuate our estimated effect. We find no evidence of this.

III.B. Procedures

On arrival at the lab, each subject is seated at a carrel with a mug on it. We then flip a coin in front of each subject individually. The coin's sides are labeled 1 and 8, and we give each subject an index card with their resulting number on it. Subjects then start reading instructions on the computer screen in front of them. Subjects are told that if a 10-sided die that we roll individually for each subject at the end of the study comes up lower than the number on their index card, they will receive the mug in front of them for free and leave with it at the end of the experiment.²¹ They are also told that if the die comes up 9, they will have the option to keep the mug or exchange it for a randomly determined amount of money between \$0 and \$10.

Thus, subjects whose coin came up 1 have a 10% chance of receiving the mug without making a choice (we refer to them as being in *Treatment M^L*, for low probability of getting the mug) whereas subjects who got an 8 have an 80% chance of receiving the mug without making a choice (*Treatment M^H*). Subjects in both treatments have a 10% chance of having a choice between the mug and money.

The next phase of the experiment is then identical to Experiment 1. The experimenter reads the most important parts of the instructions out loud; subjects fill out a personality questionnaire in two parts and get reminded in the middle that they may get the mug for free or have the possibility to choose between the mug and money. They are then told that they will now make choices (on a list) between different monetary amounts or keeping the mug (and not getting the money), and that, if the die comes up 9, they will receive their choice from one randomly selected row. Subjects then make their choices for dollar amounts ranging from \$0 to \$9.57, in increments of \$0.33.²² They are also told that their decisions will only be revealed to the experimenters in case the die comes up such that one of their choices applies.

Once they have made their choices for the case in which the die comes up 9, subjects get presented with a page of instructions that informs them that if the die comes up 8 (in which case they previously expected to get neither the mug nor money), they will

21. While the 10-sided die we used for Experiment 1 had numbers from 1 to 10, the die we used for this study had numbers from 0 to 9.

22. The increments were chosen to avoid focal numbers as much as possible. The software enforced a single switching point and required that subjects make a choice in all the rows. For more details, see the Online Appendix.

get their choice between a pen and a randomly determined dollar amount. It is made very clear to the subjects that their choice for this contingency does not in any way influence what they will get in case the die does not come up 8. They are handed a pen to inspect and are then again asked to make choices between keeping the pen and different dollar amounts (\$0 to \$9.57, in \$0.33 increments).

The experiments were again run at the Harvard Decision Science Laboratory. A total of 112 subjects (66 females; mean age 22), all of them either undergraduate (83 subjects) or graduate (29 subjects) students at Harvard, participated. We conducted 16 sessions with between 3 and 11 subjects per session, across five days in April and May 2010. Subjects received a show-up fee of \$10, and the experiment took about 20 minutes.²³

III.C. Results

RESULT 2. Subjects who have a high chance of leaving with the mug value the mug significantly higher than subjects who have a low chance of leaving with the mug. We estimate that being in treatment M^H instead of M^L increases willingness-to-accept for the mug by 20–30%.

Based on the individual coin flips, 52 subjects were in treatment M^H and 60 subjects in treatment M^L . We start by simply comparing the mean WTA for the mug across treatments.²⁴ On average, subjects in the M^H treatment request \$4.12 to give up the mug, while the mean WTA of subjects in the M^L treatment is \$3.74. Although these averages are in the right direction to support our theoretical predictions, they are far from statistically significantly different ($p = .44$, t -test²⁵) due to the large

23. As in Experiment 1, the subjects afterward participated in a second, completely unrelated pilot experiment, and the fact that these experiments were unrelated was made very clear to subjects.

24. In our main analysis, we measure WTA as the midpoint between the highest amount for which the subject prefers keeping the item and the lowest amount at which she prefers returning the item in exchange for the money. For subjects who prefer keeping the item for all amounts, we set their WTA to \$9.57 (the highest amount we offer them). We control for this censoring in some of our regressions (by using tobit or interval regressions) and find that it does not materially affect our results.

25. All tests reported in this article are two-sided. Though in this section we report t -tests (assuming unequal variances), nonparametric Fisher–Pitman permutation tests in all cases give very similar results, often with slightly smaller

between-subjects variation (the standard deviation of WTA_{mug} is around 2.5 in both treatments).

However, as explained in the second part of Section III.A, we also elicit WTA for the pen, WTA_{pen} , to attempt to control for individual-specific variation in WTA for university memorabilia. Looking at subject-level differences between the amounts people are willing to accept in exchange for the mug or the pen, $WTA_{mug} - WTA_{pen}$, the treatment effect indeed moves closer toward significance: on average, subjects in the M^H treatment require \$0.92 more to give up the mug than to give up the pen, while the corresponding average for the M^L treatment is only \$0.02. This treatment difference is borderline significant ($p = .058$, t -test).²⁶

Differences in absolute values between WTA for the mug and the pen are, however, not the correct functional form according to the theory. As described in Section III.A, the effect of a high probability of getting the mug should be approximately proportional to the consumption utility of the mug, or, as we implement it, linear in logs. The mean $\ln(WTA_{mug})$ is 1.30 in Treatment M^H and 1.11 in Treatment M^L ; this difference is not quite statistically significant ($p = .17$, t -test). However, when we consider the subject-level difference between $\ln(WTA_{mug})$ and $\ln(WTA_{pen})$, we find a mean of 0.33 for subjects in the M^H treatment and 0.01 for subjects in the M^L treatment. This difference is statistically significant ($p = .03$, t -test).²⁷ Thus, controlling for idiosyncratic variation in WTA enables us to reduce noise and detect the treatment effect.

Regression analysis allows for a more flexible relationship between $\ln(WTA_{mug})$ and $\ln(WTA_{pen})$ and allows us to control for demographic characteristics and potential day-specific effects that may influence individuals' preference for the mug versus the pen. It can also address the censoring that occurs in a few cases at the top or the bottom of the WTA scale. Our main regression equation is

$$\ln(WTA_{mug,i}) = \alpha + \beta \cdot M_i^H + \gamma \ln(WTA_{pen,i}) + \psi' X_i + \epsilon_i,$$

p -values. (The Fisher–Pitman test is a more powerful alternative to the better known Wilcoxon–Mann–Whitney test; see Kaiser [2007] for further discussion.)

26. The average WTA for the pen, WTA_{pen} , is somewhat higher in the M^L treatment (\$3.72) than in the M^H treatment (\$3.20), but the means are not statistically significantly different ($p = .27$, t -test; $p = .61$ if instead we use $\ln(WTA_{pen})$).

27. The nonparametric Fisher–Pitman test yields $p = .02$; the less powerful Wilcoxon–Mann–Whitney test $p = .09$. Subjects that indicate a \$0 WTA for either the mug or the pen, meaning they would rather have nothing than leave with the item, get dropped when we use $\ln(WTA)$. There are five such subjects in M^H (two of which indicate a \$0 WTA for both items) and three in M^L .

where M_i^H is an indicator that equals 1 for subjects in treatment M^H . If expecting to get the mug with a high probability increases WTA for the mug, as predicted by the KR theory, we expect $\beta > 0$. If instead preferences are reference-independent, or if reference points are given by the status quo, we expect $\beta = 0$.

Table II continues to show support for expectation-based reference points. Column (1), where we regress $\ln(WTA_{mug,i})$ on the treatment indicator M_i^H only, restates the result that without controlling for WTA_{pen} , β is positive but not quite statistically significant. However, as predicted in the discussion in Section III.A, once we add $\ln(WTA_{pen,i})$ as an additional regressor to reduce idiosyncratic noise, β increases in magnitude to 0.27 and becomes significant at $p < .05$ (column (2)). It remains significant and increases even further in magnitude (to 0.3) if we add demographic and day controls (column (3); note that the signs of the coefficients on age and female are consistent with what was found in Experiment 1) or use a tobit that accounts for the few censored observations (at $WTA_{mug} = 9.57$; column (4)). In column (5), we allow for a more flexible relationship between $\ln(WTA_{mug,i})$ and $WTA_{pen,i}$ by using a cubic function in $WTA_{pen,i}$ instead of $\ln(WTA_{pen,i})$. This also enables us to include the two subjects with $WTA_{pen,i} = 0$ in the sample. The point estimate of the treatment effect decreases slightly but remains significant at $p < .05$.

TABLE II
DETERMINANTS OF WILLINGNESS-TO-ACCEPT FOR THE MUG

	(1)	(2)	(3)	(4)	(5)
Treatment M^H	0.194 (0.143)	0.266** (0.124)	0.306** (0.128)	0.308** (0.126)	0.290** (0.13)
$\ln(WTA_{pen})$		0.524*** (0.091)	0.560*** (0.093)	0.568*** (0.091)	
Female			-0.199 (0.131)	-0.213* (0.128)	-0.191 (0.133)
Age			-0.052* (0.027)	-0.050* (0.026)	-0.046* (0.027)
Day indicators	No	No	Yes	Yes	Yes
Other controls					Cubic in WTA_{pen}
# observations	106	104	104	104	106

Notes. Dependent variable: $\ln(WTA_{mug})$. (1), (2), (3), and (5) are OLS regressions; (4) displays predicted marginal effects from a tobit that takes into account censoring at $WTA_{mug} = 9.57$. All regressions contain a constant, and except for (1) also an indicator for $WTA_{pen} = 9.57$ (which indicates that a subject's WTA for the pen may be censored; $p > .30$ in all regressions). Regressions with $\ln(WTA_{pen})$ as explanatory variable drop two subjects with $WTA_{pen} = 0$. Standard errors in parentheses.

Level of significance: * $p < .10$, ** $p < .05$, *** $p < .01$

Taken together, these results indicate that WTA for the mug increases by 20–30% when subjects have a high probability of getting the mug. In Table A.1 in the Appendix, we conduct a number of additional robustness checks that show that our result is robust to adding an interaction term $M_i^H \cdot \ln(WTA_{pen,i})$, session fixed effects, or to the use of an interval regression that accounts for the fact that we only observe bounds on subjects' WTA. Also, we show that if we run regressions in levels rather than in logs, the estimated treatment effect is generally borderline significant (at $p < .10$, and at $p < .05$ if we use interval regression) once WTA_{pen} and demographic characteristics are added as regressors.

III.D. Gain-Loss Utility or Motivated Taste Change?

The results from both our experiments show that individuals display a reluctance to give up items they expect to own. Our findings are predicted by the theory of expectation-based reference points formalized by KR, in which expectations affect individuals' gain-loss utility. However, expectations could also directly affect perceived consumption utility, that is, how desirable one finds an item. Following Strahilevitz and Loewenstein (1998), we refer to this theory as “motivated taste change.”²⁸ This theory presumes that preferences are to some extent (consciously or subconsciously) “manipulated” to maximize a utility function that may include things such as self-image, which could in part be determined by the self-perceived desirability of one's possessions. One could formalize such a theory as saying that the perceived consumption utility from an item i that an individual will own with probability p_i is given by $u_i(p_i)$, with $\frac{du_i}{dp_i} > 0$.

These two types of theories are observationally equivalent in our experiments, as well as in many other economically relevant settings involving valuation of goods. However, the KR

28. Strahilevitz and Loewenstein also attempt to test whether their findings of an effect of ownership on valuation are driven by shifting reference points or motivated taste change and argue that the former theory provides a better explanation of their findings. Using subjective attractiveness questionnaires, they find some evidence for motivated taste change for goods owned over a long period of time (about an hour), but not when ownership had only lasted for a few minutes, as in our experiment. Some researchers in psychology who use similar questionnaires find evidence that people who expect continued ownership of an item like it better even within a few minutes (e.g., Beggan 1992; Gilbert and Ebert 2002), while others do not find similar results (e.g., Barone, Shimp, and Spratt 1997). See Rick (forthcoming) for a recent survey of related topics that also includes a discussion of neuroeconomic evidence on loss aversion.

theory makes many additional, largely untested predictions in different domains, with expectation-based reference points and loss aversion as crucial ingredients (see e.g., Kőszegi and Rabin [2007] on risky choice). If motivated taste change were to explain our experimental results, they would provide less support for the application of theories of expectation-based reference points in other domains.

In an attempt to untangle the two theories, we conduct a variation of Experiment 2 for which the two theories make distinct predictions. In this new experiment, we continue to manipulate the subjects' probability of obtaining one mug for free, p_1 . However, we now measure subjects' WTA for a *second*, identical mug in a state of the world in which they are given one mug for free. Thus, diminishing marginal utility of a second mug should not affect WTA differently across treatments.

The gain-loss utility account formalized by KR predicts that WTA for a second mug should not be affected by p_1 , because a second mug is a gain with respect to both a reference point of zero mugs or one mug (see the Online Appendix for details). In contrast, motivated taste change theory predicts that WTA for the second mug is also increased by a high p_1 . This is because in this theory, individuals' beliefs about the desirability of such mugs are affected by their probability of getting one.

The procedures for this experiment are nearly identical to the ones of our valuation experiment described in Section III.B (see the Online Appendix). The exception is that subjects are now informed that in case the die comes up 9, they will receive the mug in front of them for free, plus have the possibility to choose between a second, identical mug and a randomly determined amount of money between \$0 and \$10.

Data from 56 subjects (29 in the high-probability treatment, $\tilde{M}^H$) show no support for motivated taste change in this context. Mean WTA for the second mug (WTA_{mug2}) is slightly lower in $\tilde{M}^H$ than in the low-probability treatment $\tilde{M}^L$ (\$2.28 vs. \$2.39), as is the difference between WTA_{mug2} and WTA_{pen} (\$-0.89 vs. \$-0.71).²⁹ Although $\ln(WTA_{mug2})$ is slightly higher in the high-probability treatment (0.75 vs. 0.72), the mean subject-level

29. Unsurprisingly, WTA for a second mug is thus quite a bit lower than WTA for the first mug found in Section III.C, which averaged \$3.92 across the two treatments. One might be concerned that our finding of no treatment differences is due to many subjects having very low WTA_{mug2} ; however, only seven subjects state WTA_{mug2} below \$0.50.

difference between $\ln(WTA_{mug2})$ and $\ln(WTA_{pen})$ is lower in treatment $\tilde{M}^H$ (-0.29) than in treatment $\tilde{M}^L$ (-0.20). As a reminder, in Section III.C the log-difference between WTA for one mug and WTA for the pen was 0.32 higher in the high-probability treatment. We can reject the null hypothesis that the effect of being in the high-probability treatment on $\ln(WTA_{mug2}) - \ln(WTA_{pen})$ equals 0.32 with $p < .04$ (t -test). Similarly, the coefficient on $\tilde{M}^H$ in a regression parallel to column (2) in Table II equals -0.03 , and the null hypothesis that this coefficient equals 0.266 (the treatment effect of M^H estimated in Experiment 2) is rejected at the 10% level.

Thus, the results of this experiment provide no evidence in favor of the hypothesis that a higher probability of ending up with a mug increases subjects' perceived consumption utility from the mug: the estimated treatment effect mostly goes in the opposite direction.³⁰ This evidence suggests that the findings in our main experiments were driven by gain-loss utility around expectation-based reference points, not motivated taste change.

IV. CONCLUSION

In two simple experiments, increasing subjects' expectations of leaving with an item has an effect on exchange behavior (they are more likely to choose to keep the item) and valuation (they demand more compensation to give up the item). We thus provide evidence that individuals' reference points are determined, at least in part, by expectations.

Our findings support the assumption of expectation-based reference points made by [Kőszegi and Rabin \(2006\)](#) and subsequent applications of the theory. They also contribute to a recent debate regarding the existence and interpretation of endowment effects in exchange and valuation experiments. In particular, unlike earlier work ([Knetsch 1989](#); [Kahneman, Knetsch, and Thaler 1990](#)), [Plott and Zeiler \(2005, 2007\)](#) find no endowment effect under their preferred experimental procedures. However, in their

30. A similar experiment is conducted by [Morewedge et al. \(2009\)](#), who report that people who already received a mug for free have a significantly higher WTP for a second mug than nonowners do for a first mug (or per mug if given the opportunity to buy two), consistent with the idea that owners like the mug better. The differential findings could be due to differences in how subjects are randomized into treatments, or to the fact that in [Morewedge et al.](#)'s setting ownership was certain, not probabilistic.

experiments, procedural details may have created a perception in subjects that they will be able to trade their endowed item for an alternative item or money, or that they might also be given the alternative item. Most discussions implicitly assume that reference points are given by current endowments, in which case these perceptions should not matter for behavior; however, with expectation-based reference points, they will. As a consequence, although Plott and Zeiler's findings can be interpreted as evidence against the view that the utility function is kinked at the current endowment, they are not incompatible with a kink around *expected* outcomes. Indeed, our experiments provide support for this view and suggest that an "endowment effect" of some sort is real, but operates via expectations instead of formal ownership.³¹

The implications of expectation-based reference points for policy are often similar to those of status quo-based reference points, since ownership often carries with it the expectation of keeping a good. Yet there are consequential differences. For instance, loss aversion may not inhibit trade in markets where individuals expect to trade, such as markets for tradeable pollution (e.g., carbon) permits.

Though we show that expectations are an important determinant of reference points, we cannot rule out that other factors, such as social norms, aspirations, salience, and history, may also influence the reference point. Untangling these factors suggests an interesting direction for future research.

APPENDIX

A. *Summary of the Kőszegi and Rabin (2006) Theory*

KR (2006) extend [Kahneman and Tversky's \(1979\)](#) prospect theory and bring it closer to "standard" economic theory. In their formulation, the utility of a decision maker (henceforth DM) depends both on her consumption c and on how this consumption compares with a reference level r . KR assume separability between "consumption utility" $m(c)$ and "gain-loss utility" $n(c|r)$. Consumption and reference levels can have multiple dimensions, so that $c = (c_1, c_2, \dots, c_K)$ and $r = (r_1, r_2, \dots, r_K)$, and it is assumed that both forms of utility are additively separable across dimensions, such that $m(c) = \sum_{k=1}^K m_k(c_k)$ and $n(c|r) = \sum_{k=1}^K n_k(c_k|r_k)$. Total

31. We discuss the relationship between our results and previous exchange and valuation experiments in more detail in the working paper version of this article.

reference-dependent utility (RDU) of an outcome c given reference level r is then given by

$$u(c|r) = \underbrace{\sum_{k=1}^K m_k(c_k)}_{\text{cons. utility } m(c)} + \underbrace{\sum_{k=1}^K \mu(m_k(c_k) - m_k(r_k))}_{\text{gain-loss utility } n(c|r)},$$

where $\mu(\cdot)$ is continuous and strictly increasing and has the properties of the Kahneman-Tversky value function: loss aversion (a kink at 0) and diminishing sensitivity. In our derivations, we use the piecewise linear specification (as in Section IV of KR 2006) $\mu(x) = \eta x$ for $x > 0$ and $\mu(x) = \eta\lambda x$ for $x \leq 0$, where $\eta > 0$ is the DM's weight on gain-loss utility and $\lambda > 1$ her loss aversion coefficient.

In KR, the reference point is given by probabilistic beliefs about future outcomes, so that it is in fact a reference lottery given by probability measure $G : \mathbb{R}^K \rightarrow \mathbb{R}$ over consumption utility in each of the k dimensions. Thus the RDU of outcome c given expectations G is

$$U(c|G) = \int u(c|r) dG(r).$$

This formulation implies that when evaluating a dimension c_k of a consumption outcome c , the DM compares it separately with each possible value that this dimension could take according to the reference lottery and weights it by the probability of this outcome in the reference lottery.

But what exactly determines the reference lottery? KR employ the concept of *personal equilibrium* (PE), which requires the following:

- the DM has a plan for every contingency that she can possibly face in a given decision situation;
- her reference lottery is based on her expectation to put these plans into action once she knows which contingency applies; and
- her actions maximize expected RDU, given her reference lottery.

The timing of a decision situation in which PE applies is as follows:

- $t = -1$: DM starts focusing on the decision; knows all the possible choice sets she might face at $t = 1$ and the probabilities with which they occur;

- $t = 0$: DM makes plans for each possible choice set, thereby setting her reference lottery;
- $t = 1$: “Nature” determines the choice set the DM actually faces. The DM follows through on her plan for this choice set, which then yields either a deterministic or a probabilistic outcome.

The PE concept requires the DM’s actions at $t = 1$ to be optimal given her beliefs at $t = 0$, beliefs formed at $t = 0$ to be rational given her plans, and her plans to be “credible” in the sense that at $t = 1$ the DM has no incentive to deviate from them (akin to subgame perfection in standard multiplayer game theory).³² In our experiments, we elicit decisions right before “nature” (the die roll) determines which choice set actually applies. This does not alter the theoretical predictions, because of the consistency requirement imposed by the PE concept: once a subject has made her plans (which determine her reference lottery), it does not matter whether we elicit the plans or actual decisions, because in PE the DM would not deviate from her plans once she knows which choice set she actually faces.

KR emphasize that it is possible for multiple PEs to exist in a given situation. An example: assume the consumption utilities resulting from actions x and y are not too different. Then one equilibrium may be for the DM to plan on x and indeed do x , but another one may be for her to plan on y and do y (because if she planned on y and ended up doing x , she would feel a loss from not getting the consequence of y , which may outweigh a possible consumption utility advantage of x). In such a case KR’s preferred way of equilibrium selection is to take the PE that gives the DM the highest expected RDU, the *preferred personal equilibrium* (PPE).

Detailed derivations of the KR predictions for our experiments are presented in the Online Appendix.

B. Additional Empirical Results for Experiment 2

Appendix Table A.1 contains multiple robustness checks of the results discussed in Section III.C. Regressions (1) to (3) use $\ln(WTA_{mug})$ as the dependent variable, as in the analysis in the main text, whereas regressions (4) to (9) instead use WTA_{mug} in levels.

32. As beliefs affect payoffs in this setup, the KR model can be seen as an example of a dynamic psychological game (Battigalli and Dufwenberg 2009).

TABLE A.1
DETERMINANTS OF WILLINGNESS-TO-ACCEPT FOR THE MUG: ROBUSTNESS

	Dependent variable: $\ln(WTA_{mug})$			Dependent variable: WTA_{mug}					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treatment M^H	0.559** (0.228)	0.37*** (0.134)	0.297** (0.13)	0.596 (0.496)	0.765* (0.426)	0.81* (0.419)	0.777* (0.448)	0.774* (0.425)	0.814** (0.413)
$\ln(WTA_{pen})$	0.616*** (0.115)	0.438*** (0.095)	0.531*** (0.093)						
Tr. $M^H \times \ln(WTA_{pen})$	-0.252 (0.179)								
WTA_{pen}					0.536*** (0.099)	0.577*** (0.098)	0.571*** (0.104)	0.547*** (0.098)	0.587*** (0.096)
Female						-0.759* (0.432)			-0.796* (0.426)
Age						-0.16* (0.087)			-0.156* (0.086)
# observations	104	104	104	112	112	112	112	112	112

Notes. All regressions other than (2) (which uses session indicators) also contain indicators for the different days on which sessions were run (none of which ever reach a significance level $p < .2$) and an indicator for $WTA_{pen} = 9.57$ (which indicates that a subject's WTA for the pen may be censored; $p > .3$ in all regressions). Standard errors in parentheses.
Level of significance: * $p < .10$, ** $p < .05$, *** $p < .01$

In regression (1), we add an interaction term $M_i^H \cdot \ln(WTA_{pen,i})$ to our main regression to allow for the possibility that the relation between $\ln(WTA_{mug})$ and $\ln(WTA_{pen})$ differs across treatments. The coefficient on this interaction term is negative but not significant. Regression (2) uses session dummies instead of date dummies as in the main text, which increases the estimate of the treatment effect somewhat. Column (3) shows the results from an interval regression that explicitly allows for the fact that we only know bounds on subjects' WTA.³³ The treatment effect remains significant at $p < .05$, with a nearly unchanged point estimate compared with the analysis in the text.

Regression (4) shows that simply comparing WTA_{mug} across treatments does not yield a significant treatment effect. However, once WTA_{pen} is added as a regressor, the estimated treatment effect increases in magnitude, as in the log case, and is significant at $p < .10$ (column (5)). Column (6) adds demographic controls, and column (7) displays the marginal effects from a tobit regression that takes into account censoring at $WTA_{mug}=0$ and $WTA_{mug}=9.57$. Columns (8) and (9) display results from interval regressions and show that once demographic characteristics are added as regressors, the treatment effect is significant at $p < .05$. Overall, we conclude that the results from estimating our main equation (in logs) are robust to a variety of different regression specifications.

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SUPPLEMENTARY MATERIAL

An Online Appendix for this article can be found at QJE online (qje.oxfordjournals.org).

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